

LCG - BEII - Introduction and contents

LCG-BEII 2019

Jacques van Helden

2025-02-04

Scope of the course

This course provides an overview of bioinformatics methods used to analyse cis-regulatory elements in non-coding DNA sequences.

1. Detection of cis-regulatory elements and other types of nucleic sequence motifs.
2. Next Generation Sequencing approaches to genomic regulation
 - ▶ ChIP-seq for transcription factors
 - ▶ ChIP-seq for chromatin marks (histone acetylation, methylation)
 - ▶ RNA-seq (differentially expressed genes)
3. Underlying statistics

Schedule

Day	Topics
Day 1	Regulatory sequences: cis-regulatory elements intro + predicting binding sites
Day 2	Regulatory sequences: motif discovery
Day 3	Analysing regulation with NGS data
Day 4	Integrative exercises and evaluations

General goals of the course

- ▶ Learn the **concepts and methods** related to bioinformatics analysis of cis-regulation
- ▶ Understand the underlying **algorithms** and **statistics**.
- ▶ Get a **practical experience** of bioinformatics resources (**databases** and **tools**) used in the domain

Specific goal: skills in statistics

1. Topics

- ▶ Sequence models
- ▶ Statistics of motif detection (scanning, discovery)
- ▶ NGS statistics (read mapping, peak calling, differential expression)
- ▶ Designing control sets to validate the statistical models

2. Approach

- ▶ Minimalist theoretical elements
- ▶ Intuitive approach to the statistics: lots of graphs, simulations, controls

3. Environment

- ▶ R statistical language (<https://www.r-project.org/>)
- ▶ Bioconductor: (<https://www.bioconductor.org/>)
- ▶ Developing environment: RStudio (<https://www.rstudio.com/>)

Specific goal: technical skills

- ▶ Writing reports in R markdown (<http://rmarkdown.rstudio.com/>)
- ▶ Version management with git (<https://git-scm.com/>) - see Blischak, Davenport, and Wilson (2016)

Resources for this course

Name	Description + URL
LCG_BEII	The course material for this course on github http://jvanheld.github.io/LCG_BEII
RSAT	Regulatory Sequence Analysis Tools http://rsat.eu/
RSAT course	Teaching material for the analysis of regulatory sequences http://rsa-tools.github.io/course/

Downloading a local copy of the course material

This is optional.

If you wish to download a local copy of all the course material, open a Unix terminal and type.

```
git clone https://github.com/jvanheld/LCG_BEII.git
```

Then open the file *LCG_BEII/index.html* with a Web browser.

Since the course will be periodically improved, you can get the updates with the following command.

```
cd LCG_BEII;  
git pull
```

Command-line use of RSAT for LCG students

LCG students can access RSAT on the command line with the following commands

```
## Open an ssh connection to the local server  
ssh tepeu
```

```
## Load the RSAT configuration file  
source /export/apps/rsat/RSAT_config.bashrc
```

```
## Check that the paths are correct  
supported-organisms
```

References

Blischak, John D, Emily R Davenport, and Greg Wilson. 2016. "A Quick Introduction to Version Control with Git and GitHub." *PLoS Computational Biology* 12 (1): e1004668.